# Final Project Report

## AI-Powered Phishing Email Analyzer with Automated Incident Reporting

Submitted By: Muhammad Sameer | Internship Role: AI, Automation & Security Engineering Intern | Organization: The Arzens Pvt. Ltd.

### 1. Project Overview

This project is a web-based cybersecurity tool that analyzes suspicious email content and helps users understand whether an email is safe or potentially phishing. The system checks the sender, subject, email body, links, and attachment name, then calculates a phishing score, classifies the risk level, explains the detected indicators, recommends actions, saves the record, and generates an incident report.

### 2. Problem Statement

Phishing emails are common and often look professional. A normal user may not notice fake login pages, urgent wording, suspicious links, sender impersonation, or risky attachments. Manual checking also takes time. This project reduces that effort by providing a quick automated analysis and report.

### 3. Objectives

The objective is to analyze suspicious email text, detect phishing indicators, classify risk as Low, Medium, High, or Critical, generate a clear explanation, recommend security actions, and create an automated incident report for documentation.

### 4. System Workflow

The workflow is simple: the user submits email details through a web form, the Python backend sends the data to the analyzer, the analyzer calculates a score and risk level, the result is saved in SQLite, and the system generates an HTML incident report that can be opened from the dashboard.

### 5. Main Features

- Email content input form
- Sender, subject, body, link, and attachment analysis
- Phishing score from 0 to 100
- Risk level classification: Low, Medium, High, Critical
- Detected phishing indicators and explanation
- Recommended security actions
- SQLite dashboard for previous reports
- HTML incident report generation
- CSV export for analysis history
- Optional n8n webhook alert for high-risk emails

## 6. Technology Stack

| Component | Technology Used |
|---|---|
| Frontend | HTML, CSS, JavaScript |
| Backend | Python standard library HTTP server |
| Database | SQLite |
| Analysis Engine | Local phishing risk scoring and AI-style explanation logic |
| Reports | HTML incident reports |
| Automation | Optional n8n webhook alert |

## 7. Risk Scoring Logic

The analyzer checks phishing signals such as urgent words, password or login requests, risky actions, financial terms, non-HTTPS links, URL shorteners, IP-based links, sender lookalike patterns, free email domains used for brand-related messages, and risky attachment extensions. Based on these indicators, it calculates a score and maps it to a risk level.

## 8. Database and Reporting

Every analysis result is saved in a local SQLite database. The dashboard shows the report history with sender, subject, risk level, score, and view option. The incident report contains email details, risk score, indicators, explanation, and recommended actions.

## 9. Security Value

This project is useful for security awareness and basic SOC triage. It helps users quickly identify suspicious emails, document incidents, and take safer actions before interacting with links or attachments.

## 10. Limitations and Future Work

The current version uses a local analysis engine and does not replace a professional security investigation. In future versions, real LLM API integration, Gmail inbox scanning, Google Sheets logging, PDF export, login system, and bulk email scanning can be added.

## 11. Conclusion

The final project demonstrates the practical use of AI, cybersecurity, and automation in a simple web-based tool. It provides a complete flow from suspicious email input to risk classification, recommendations, database storage, dashboard view, and incident reporting.